Subject Code	:	CEUE203	Semester	:	3	Academic Year	:	2026-27
Subject Name	:	Object-Oriented Programming						

	Practical 2
Program: 2.1	**Smart thermostat (Thermostat.java, main in this file)** (a) Create class Thermostat with private String location, private int temperature, private static final int MIN=16 and MAX=30, and private static int activeCount=0. (b) Constructor(location, startTemp): store location; set temperature to startTemp if it is within MIN..MAX else 22; increment activeCount. CO1 PO1, PO2, PO3 Practical Number CO/PO (c) Constructor(location): chain with this(location, 22). (d) raise(): if temperature < MAX add 1, else print "Already at maximum (30)". (e) lower(): if temperature > MIN subtract 1, else print "Already at minimum (16)". (f) Add getTemperature() (no setter) and static getActiveCount(). (g) In main: create two thermostats, call raise() 10 times then lower() 20 times in loops (printing the temperature each time), then print getActiveCount(). Expected output: temperature rises to 30 then "Already at maximum", falls to 16 then "Already at minimum", count 2.

Code	class Thermostat
	{
	private String location;
	private int temperature;
	private static final int MIN=16,MAX=30;
	private static int ActiveCount=0;
	Thermostat(String location,int temperature)
	{
	if(temperature>=MIN && temperature<=MAX)
	{
	this.location=location;
	this.temperature=temperature;
	ActiveCount++;
	}
	else
	{
	this.location=location;
	this.temperature=22;
	ActiveCount++;
	}
	}
	Thermostat(String location)
	{
	this(location,22);
	}
	public int getTemperature()
	{

```
        return temperature;
    }
    public String getLocation()
    {
        return location;
    }
    public static int getActiveCount()
    {
        return ActiveCount;
    }
    public void raise()
    {
        if(temperature<MAX)
        {
            temperature+=1;
        }
        else
        {
            System.out.println("Already at maximum (30)");
        }
    }
    public void lower()
    {
        if(temperature>MIN)
        {
            temperature-=1;
        }
        else
        {
            System.out.println("Already at minimum (16)");
        }
    }
}
public class Thermostat_prac
{
    public static void main(String[] args)
    {
        Thermostat t1=new Thermostat("hall");
        Thermostat t2=new Thermostat("bedroom",25);
        int i;
        for(i=0;i<10;i++)
        {
            t1.raise();
            int temp=t1.getTemperature();
            String loc=t1.getLocation();
            System.out.print("Location : "+ loc);
            System.out.println("   temperature : "+ temp);
        }
        for(i=0;i<20;i++)
        {
            t1.lower();
            int temp=t1.getTemperature();
```

```
            String loc=t1.getLocation();
            System.out.print("Location : "+ loc);
            System.out.println("   temperature : "+ temp);
        }
        for(i=0;i<5;i++)
        {
            t2.raise();
            int temp=t2.getTemperature();
            String loc=t2.getLocation();
            System.out.print("Location : "+ loc);
            System.out.println("   temperature : "+ temp);
        }
        int cnt=Thermostat.getActiveCount();
        System.out.println("active count : "+ cnt);
    }
}
```

Output

```
C:\materials\3rd sem\OOP java\practicals\practical 2>javac Thermostat_prac.java

C:\materials\3rd sem\OOP java\practicals\practical 2>java Thermostat_prac
Location : hall   temperature : 23
Location : hall   temperature : 24
Location : hall   temperature : 25
Location : hall   temperature : 26
Location : hall   temperature : 27
Location : hall   temperature : 28
Location : hall   temperature : 29
Location : hall   temperature : 30
Already at maximum (30)
Location : hall   temperature : 30
Already at maximum (30)
Location : hall   temperature : 30
Location : hall   temperature : 29
Location : hall   temperature : 28
Location : hall   temperature : 27
Location : hall   temperature : 26
Location : hall   temperature : 25
Location : hall   temperature : 24
Location : hall   temperature : 23
Location : hall   temperature : 22
Location : hall   temperature : 21
Location : hall   temperature : 20
Location : hall   temperature : 19
Location : hall   temperature : 18
Location : hall   temperature : 17
Location : hall   temperature : 16
Already at minimum (16)
Location : hall   temperature : 16
Already at minimum (16)
Location : hall   temperature : 16
Already at minimum (16)
Location : hall   temperature : 16
Already at minimum (16)
Location : hall   temperature : 16
Already at minimum (16)
Location : hall   temperature : 16
Already at minimum (16)
Location : hall   temperature : 16
Location : bedroom   temperature : 26
Location : bedroom   temperature : 27
Location : bedroom   temperature : 28
Location : bedroom   temperature : 29
Location : bedroom   temperature : 30
active count : 2
```

Program: 2.2	**Cinema show (CinemaShow.java, main in this file)** (a) Create class CinemaShow with private String title, private int seatsAvailable, private final int capacity, and private static int totalBooked=0. (b) Constructor(title, capacity): set both and seatsAvailable = capacity. Constructor(title): chain with this(title, 100). (c) book(int n): if n ≤ seatsAvailable, reduce seatsAvailable by n, add n to totalBooked, return true; else return false unchanged. (d) cancel(int n): increase seatsAvailable by n but never above capacity. (e) Add getSeatsAvailable() and static getTotalBooked(). (f) In main: run a sequence of book and cancel calls, printing the result and seats after each, then print getTotalBooked(). Expected output: booking more than available prints false and leaves seats unchanged; totalBooked equals the successful bookings.

Code:	class Cinema
	{
	private String title;
	private int seatsAvailable;
	private final int capacity;
	private static int totalBooked=0;
	Cinema(String title,int capacity)
	{
	this.title=title;
	this.capacity=capacity;
	this.seatsAvailable=capacity;
	}
	Cinema(String title)
	{
	this(title,100);
	}
	public boolean book(int n)
	{
	if(seatsAvailable>=n)
	{
	seatsAvailable-=n;
	totalBooked+=n;
	return true;
	}
	else
	{
	return false;
	}
	}
	public void cancel(int n)
	{
	if(seatsAvailable+n<=capacity)
	{
	seatsAvailable+=n;
	
	}
	else
	{
	seatsAvailable=capacity;
	}
	}

```
    public int getSeatsAvailable()
    {
        return seatsAvailable;
    }
    public static int getTotalBooked()
    {
        return totalBooked;
    }
    public void display()
    {
        System.out.println("title :"+title);
        System.out.println("seats available :"+getSeatsAvailable());
        System.out.println("total booked :"+getTotalBooked());
    }
}
public class cinema_prac
{
    public static void main(String[] args)
    {
        Cinema c1=new Cinema("movie1",100);
        Cinema c2=new Cinema("movie2",200);

        boolean result=c1.book(50);
        System.out.println("booking result :"+result);
        c1.display();
        c1.cancel(20);
        c1.display();

        result = c1.book(100);
        System.out.println("booking result : " + result);
        c1.display();

        result=c2.book(20);
        System.out.println("booking result :"+result);
        c2.display();
        c2.cancel(30);
        c2.display();

        System.out.println("Final Total Booked : " + Cinema.getTotalBooked());
    }
}
```

Output

```
C:\materials\3rd sem\OOP java\practicals\practical 2>javac cinema_prac.java

C:\materials\3rd sem\OOP java\practicals\practical 2>java cinema_prac
booking result :true
title :movie1
seats available :50
total booked :50
title :movie1
seats available :70
total booked :50
booking result : false
title :movie1
seats available :70
total booked :50
booking result :true
title :movie2
seats available :180
total booked :70
title :movie2
seats available :200
total booked :70
Final Total Booked : 70
```